

Muhammad Ahmad Sultan, M.S.

Ph.D. Candidate (4th Year), Electrical & Biomedical Engineering

+1-614-496-1819 | sultan.47@osu.edu

[in linkedin/ahmad/](https://www.linkedin.com/in/ahmad/) | [website/ahmad/](https://www.muhammadahmadsultan.com/)

Columbus, OH - 43201, USA

EDUCATION

- **Ph.D. Biomedical Engineering | Post-candidacy** May 2027
The Ohio State University, Columbus, OH, USA (Expected)
 - GPA: 3.90/4.00
 - Advisor: Rizwan Ahmad, Ph.D., Lab: Cardiovascular Magnetic Resonance Lab [🌐]
 - **Research Focus:** Development of **self-supervised deep learning** methods for accelerated cardiac MRI reconstruction to improve image quality, reduce scan time, and streamline acquisition workflows.
- **M.S. Biomedical Engineering (Machine Learning)** Dec 2024
The Ohio State University, Columbus, OH, USA
 - GPA: 3.90/4.00
 - **Relevant Coursework:** Reinforcement Learning, Optimization, Probability Theory, **Machine Learning** in BME, MR Spectroscopy & Imaging, Medical Imaging & Analysis
- **B.S. Electrical & Computer Engineering | Gold Medalist** May 2021
University of Engineering and Technology, Lahore, Pakistan
 - GPA: 3.90/4.00, Class Rank: 1/200
 - **Relevant Coursework:** Signal Processing, **Machine Learning**, Data Structures & Algorithms, Calculus, Linear Algebra, Embedded Systems, Digital System & Logic Design

RELEVANT EXPERIENCE

- **Graduate Research Associate @ Cardiovascular Magnetic Resonance Lab [🌐]** Aug 2022 – Present
The Ohio State University, Columbus, OH, USA
 - Developed DISCUS, a self-supervised **generative deep learning** reconstruction framework, enabling accelerated dynamic MRI reconstruction by combining **deep image prior and temporal manifold learning of image series**, demonstrating improved diagnostic image quality using NMSE & SSIM for single-shot free-breathing LGE imaging. | [Paper](#) | [Code](#)
 - Extending ML-DIP framework, a multi-dynamic low-rank **deep learning-based generative DIP**, to motion-robust **3D LGE imaging** with **explicit deformation fields** for respiratory and cardiac motion, integrating Dixon fat-water modeling for multi-echo and joint sensitivity maps estimation, trained on **multi-GPU cluster with NVIDIA H100s**, for accelerated ~5-minutes iNAV acquisition, with better arrhythmia handling. | [MLE-DIPS Abstract](#) | [Cine ML-DIP Paper](#)
- **Research Assistant @ Signal, Image & Video Processing Lab** Aug 2021 – July 2022
Lahore University of Management Sciences, Lahore, Pakistan
 - Developed a **machine learning** pipeline for a **Vitals Monitoring System** using only PPG time-series data, derived **spectro-temporal features**, applied regression for cough-less blood pressure estimation, achieved better MAE compared to state-of-the-art methods, integrated vitals in a fingertip oximeter, enabling continuous and non-invasive monitoring with one wearable sensor outside of a hospital setting. | [Paper](#)
- **Part-time Tutor @ AI Introductory Class** Apr 2022 – July 2022
Lahore University of Management Sciences, Lahore, Pakistan
 - Tutored and mentored three junior-year computer science students in **foundational AI concepts**, including agent learning, **machine vision optimization**, and deep learning basics.
- **Senior-year Research Intern @ System Simulation Lab** Feb 2021 – June 2021
University of Engineering and Technology, Lahore, Pakistan
 - Explored emerging non-volatile memories, investigated shift errors in vertical racetrack memory & VT-codes, contributed to the development of an efficient **syndrome-calculator algorithm** for VRM using radix-4 method.

SKILLS

- **Research Areas:** Accelerated Image Reconstruction, Unsupervised Deep Learning, Time-Series Modeling, Medical Imaging, Generative Modeling, Motion-Robust Imaging, Signal & Image Processing, Inverse Problems
- **Programming Languages & Data Analysis:** Python, MATLAB, C, SQL, Verilog, JMP (DOE)
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Scikit-learn, NumPy, Pandas

JOURNAL PUBLICATIONS

- C. Chen, M. Vornehm, Z. Bu, P. Chandrasekaran, **M.A. Sultan**, et al. "A multi-dynamic low-rank deep image prior (ML-DIP) for real-time 3D cardiovascular MRI." *Journal of Cardiovascular Magnetic Resonance Imaging (JCMR)*, Nov 2025. | [Paper](#)
- M. Vornehm, C. Chen, **M.A. Sultan**, et al. "Multi-dynamic deep image prior for cardiac MRI." *Magnetic Resonance in Medicine (MRM)*, 94 (6): 2668-2679, July 2025. | [Paper](#) | [Code](#)
- **M.A. Sultan**, C. Chen, et al. "An unsupervised method for MRI recovery: Deep image prior with structured sparsity." *Magnetic Resonance Materials in Physics, Biology and Medicine (MAGMA)*, 38: 859-871, May 2025. | [Paper](#) | [Code](#)
- **M.A. Sultan**, & W. Saadeh, "Continuous patient-independent estimation of respiratory rate and blood pressure using robust spectro-temporal features derived from photoplethysmogram only." *IEEE Open Journal of Engineering in Medicine and Biology (EMB)*, 5: 637-649, Nov 2023. | [Paper](#)

CONFERENCE PUBLICATIONS

- **M.A. Sultan**, C. Chen, Y. Liu, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)
- X. Lei, P. Schniter, C. Chen, **M.A. Sultan**, et al. "Surface coil intensity correction for MRI." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)
- **M.A. Sultan**, & W. Saadeh, "Robust estimation of respiratory rate from photoplethysmogram with respiration quality analysis." 2022 *IEEE International Symposium on Circuits and Systems (ISCAS)*, Austin, Texas, USA, 65-69. | [Paper](#)

PEER-REVIEWED ABSTRACTS

- **M.A. Sultan**, et al. "A motion-robust dual-echo 3D LGE reconstruction framework." Accepted for oral presentation at **SCMR 2026 Annual Scientific Sessions**, Rio de Janeiro, Brazil.
- M. Vornehm, C. Chen, **M.A. Sultan**, et al. "Motion-Guided Deep Image Prior for Dynamic Cardiac MRI." 2025 *ISMRM & ISMRT Annual Meeting & Exhibition*, Honolulu, Hawai'i, USA. | [Link](#)
- C. Chen, **M.A. Sultan**, et al. "FlowDIP: Real-time phase-contrast MRI reconstruction with flow-conditional deep image prior." *Journal of Cardiovascular Magnetic Resonance Imaging (JCMR)*, 27: 101504. Presented at **SCMR 2025 Annual Scientific Sessions**, Washington, DC. | [Link](#)
- **M.A. Sultan**, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2023 *ISMRM & ISMRT Annual Meeting & Exhibition*, Toronto, Canada. | [Link](#)

ORAL PRESENTATIONS

- **M.A. Sultan**, C. Chen, Y. Liu, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)

POSTER PRESENTATIONS

- **M.A. Sultan**. "Motion-robust 3D late gadolinium enhancement Imaging (LGE)." 2025 *Davis Heart and Lung Research Institute (DHLRI) Research Day'25*, Columbus, OH, USA.
- X. Lei, P. Schniter, C. Chen, **M.A. Sultan**, et al. "Surface coil intensity correction for MRI." 2024 *IEEE International Symposium on Biomedical Imaging (ISBI)*, Athens, Greece, 1-5. | [Paper](#)
- **M.A. Sultan**, et al. "Deep image prior with structured sparsity for dynamic MRI reconstruction." 2023 *ISMRM & ISMRT Annual Meeting & Exhibition*, Toronto, Canada. | [Link](#)
- **M.A. Sultan**, & W. Saadeh, "Robust estimation of respiratory rate from photoplethysmogram with respiration quality analysis." 2022 *IEEE International Symposium on Circuits and Systems (ISCAS)*, Austin, Texas, USA, 65-69. | [Paper](#)

INVENTIONS & PATENTS

- R. Ahmad, **M.A. Sultan**, & S.M. Arshad, "3D real-time cardiovascular magnetic resonance imaging (CMR)." Patent application in progress, OSU started filing on Sept 2025.
- R. Ahmad, **M.A. Sultan**, et al. "Systems and methods for dynamic image reconstruction." Patent pending, filed on Nov 2024, Application number: 63/716,363.

PRE-DOCTORAL FELLOWSHIPS

- **American Heart Association (AHA)**, "Motion-robust dual-echo magnetic resonance imaging-based 3D late gadolinium enhancement." Under-review, application submitted on Sept 2025.
- **The Ohio State University Presidential Fellowship**, "Accurate, Comprehensive Assessment of Heart Fibrosis Using 3D Motion-robust MRI-based Late Gadolinium Enhancement (LGE)." Under-review, application submitted on Oct 2025.

FUNDING, HONORS & AWARDS

- **NIH-funded PHD Position – Cardiovascular Magnetic Resonance Lab**  Aug 2022 - Present
The Ohio State University, Columbus, OH, USA
 - Secured a fully-funded GRA Position in CMR Lab by contributing to several advisor's grants.
- **Academic Excellence – Gold Medalist** July 2021
University of Engineering and Technology, Lahore, Pakistan
 - **Awarded 5 gold medals** for highest departmental GPA across semesters.
- **Merit Scholarship – Top 5 Students** 2017 – 2021
University of Engineering and Technology, Lahore, Pakistan
 - Dean's merit scholarship awarded to top 5 undergraduate students each semester.
- **Best Semester Project – ML Class** Dec 2020
University of Engineering and Technology, Lahore, Pakistan
 - Recognized for top project "NLP-based Fake News Detector" for machine learning class.
- **Sports Tournament – Best Player Award** Mar 2019
University of Engineering and Technology, Lahore, Pakistan
 - Best bowling performance and led the champions team in Cricket tournament, ElectroCup2k19.

UNDERGRAD SEMESTER PROJECTS

- **Fake News Detection Using Machine Learning & Natural Language Processing** Jan 2021
University of Engineering and Technology, Lahore, Pakistan
 - Used **Tf-Idf & word embeddings** for feature selection, compared performance of **Naive Bayes, SVC, ANN & LSTMs**, signified better performance using t-test with 95% confidence interval.
- **Fire & Smoke Alarm using CNN** Sept 2020
University of Engineering and Technology, Lahore, Pakistan
 - Utilized **Data Augmentation** to increase sample size, deployed **Transfer Learning** with MobileNet to enable faster training.
- **Skin Cancer Type Detection from Lesion Images** May 2020
University of Engineering and Technology, Lahore, Pakistan
 - Extended **ResNet-50 & Vgg-16** Models with Transfer Learning, achieved >85% accuracy to detect 10 types of skin cancer.

VOLUNTEER WORK

- **Poster Presentation: "Motion-robust self-supervised reconstruction framework for LGE."** Oct 2025
The Ohio State University, Columbus, OH
 - Presented a poster representing BME at the Graduate Engineering Research Symposium for undergraduate students.

CERTIFICATIONS

- **Computer Vision & NLP Projects** July 2020
 - Completed **3 projects** using keras and scikit-learn: twitter sentiment analysis (NLP), house price prediction (regression), and image classification with data augmentation & transfer learning. | [Coursera](#)
- **Deep Learning Specialization by Andrew Ng** May 2020
 - **5-course** track covering neural networks, CNNs, sequence models, hyperparameter tuning, and model optimization with hands-on programming assignments. | [Coursera](#)

WORKSHOPS & SEMINARS

- Attending **weekly seminars** in BME department at The Ohio State University. Aug 2022 – Present
- Attended **Research Enhancement Sessions** on IEEE Xplore database & publications at EE, LUMS. Jan 2022
- Participated in NVIDIA Deep Learning Workshop on **Accelerated Data Science** held at EE, LUMS. Dec 2021

ADDITIONAL INFORMATION

Languages: English, Urdu, Hindi, Punjabi

Interests & Activities: Cooking, reading scientific articles, media analysis, data storytelling (memes)

REFERENCES

1. **Rizwan Ahmad** (Ph.D. Advisor)
Professor, Biomedical Engineering Department
The Ohio State University, Columbus, OH, USA
Email: ahmad.46@osu.edu | Website: <https://u.osu.edu/ahmad/> | Scholar: <https://scholar/ahmad/>
2. **Orlando P. Simonetti** (Ph.D. Mentor)
Professor of Radiology and Internal Medicine
The Ohio State University, Columbus, OH, USA
Email: simonetti.9@osu.edu | Website: <https://u.osu.edu/simonetti/> | Scholar: <https://scholar/simonetti/>